



(ii) Calculate the period of rotation of the small cog.

period = s [2]

(iii) Show that the distance moved by point X on the chain during one full rotation of the small cog is 0.24 m.

[1]

(iv) Use the information in (b)(iii) to determine the angle through which the large cog rotates during one full rotation of the small cog.

angle = rad [2]

(c) The chain of the bicycle in (b) is moved onto a smaller cog fixed to the rear wheel. The speed of the bicycle does not change.

Explain, without calculation, the effect of this change on the angular speed of the pedals.

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 [2]

[Total: 10]

